

## CONTEMPORARY REVIEW

# Iron Dysregulation and Vascular Diseases: A Contemporary Review

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**ABSTRACT:** Iron is essential for the cardiovascular system to maintain its function. Disruption of iron balance, in the form of either iron deficiency or overload, has been increasingly recognized to be linked to the pathogenesis of cardiovascular disease. The consequences of iron dysregulation in specific vascular cell types, such as endothelial cells, vascular smooth muscle cells, and macrophages, contribute to critical processes, including inflammation, oxidative stress, and vascular remodeling. In this review, we focus on the role of iron imbalance in the pathogenesis of vascular diseases, in particular atherosclerosis and aortic aneurysm. We provide an overview of cellular and systemic iron homeostasis, explore the impact of iron dysregulation on vascular cell biology, and discuss mechanistic insights into iron imbalance-induced vascular pathology. We also summarize potential therapeutic approaches targeting iron metabolism, review ongoing clinical trials, and outline future research directions needed to translate these findings into clinical interventions for vascular disease.

**Key Words:** iron dysregulation ■ iron homeostasis ■ iron overload ■ vascular diseases

Iron is indispensable for the growth and development of all living organisms. It plays a central role in key physiological processes, including energy production, oxygen transport, and enzymatic reactions. However, unbound free iron can produce reactive oxygen species (ROS) and induce oxidative stress, which triggers cell damage and contributes to various diseases. Under physiological conditions, systemic and cellular iron metabolism is precisely regulated to maintain normal cell function.<sup>1</sup> In contrast, iron metabolism is dysregulated in response to various pathological stimuli. A better understanding of the impact of iron dysregulation on disease development and progression, and exploring the underlying mechanisms, may facilitate the development of novel strategies to improve disease management.<sup>2</sup>

Accumulating evidence suggests that iron homeostasis is essential for maintaining cardiovascular health. Iron dysregulation (iron overload and iron deficiency) can induce inflammation and oxidative stress, both of which are linked to cardiovascular diseases (CVDs).

Recently, ferroptosis, an iron-dependent form of cell death, has been recognized as an important contributor to vascular dysfunction. While growing research has focused on the role of iron in heart biology, its effects on blood vessels remain less explored. In this review, we examine how disturbances in iron homeostasis affect vascular cell function and contribute to vascular diseases, including atherosclerosis, aortic aneurysms, and peripheral artery disease (PAD). Moreover, we discuss molecular mechanisms, highlight therapeutic approaches aimed at regulating iron metabolism, and outline current clinical trial efforts.

## IRON REGULATION AND DYSFUNCTION

### Iron Regulators Maintain Iron Homeostasis

Although multiple systemic and cellular mechanisms contribute to iron homeostasis, iron regulatory proteins

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## Nonstandard Abbreviations and Acronyms

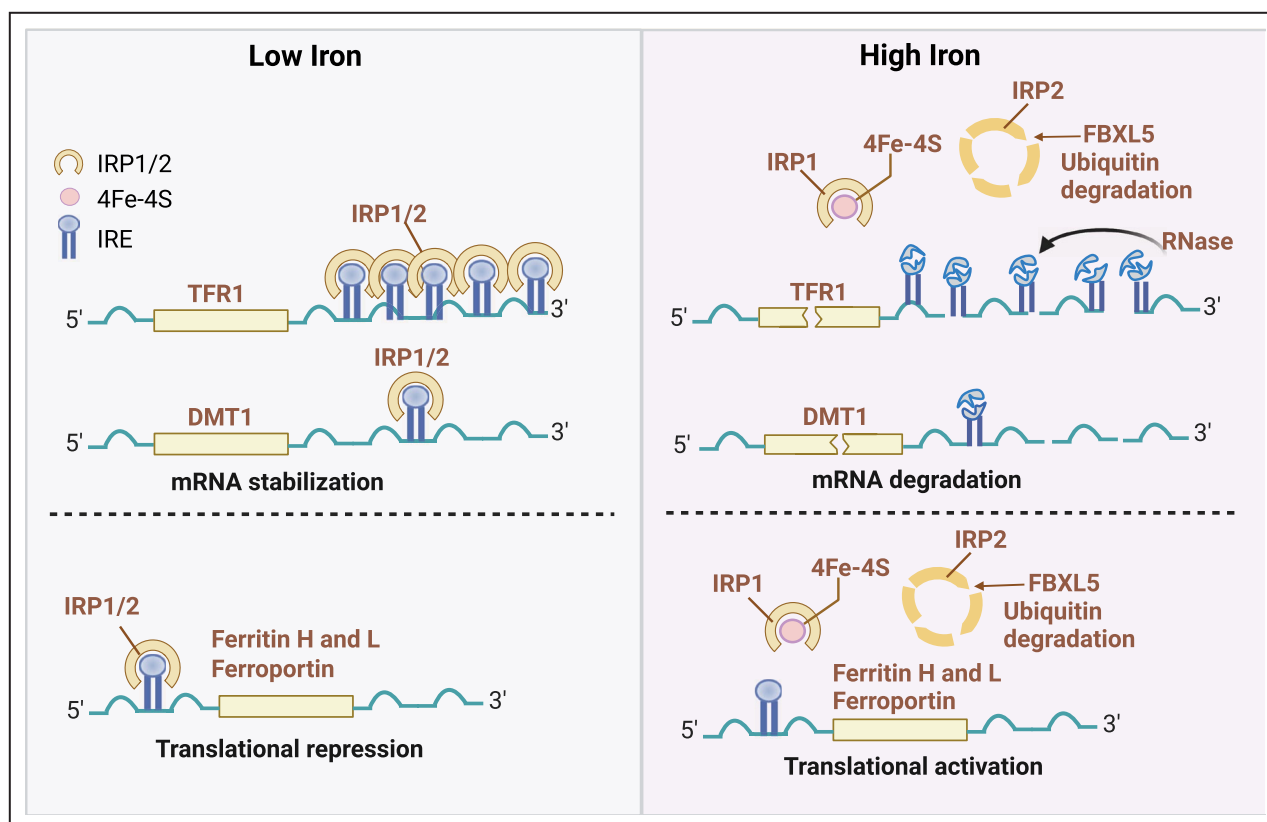
|                   |                                                                                                                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AFFIRM-AHF</b> | A Randomized, Double-Blind Placebo Controlled Trial Comparing the Effect of Intravenous Ferric Carboxymaltose on Hospitalizations and Mortality in Iron Deficient Patients Admitted for Acute Heart Failure |
| <b>CONFIRM-HF</b> | A Study to Determine the Efficacy and Safety of Finerenone and SGLT2i in Combination in Hospitalized Patients With Heart Failure                                                                            |
| <b>EC</b>         | endothelial cell                                                                                                                                                                                            |
| <b>EFFECT-HF</b>  | Effect of Ferric Carboxymaltose on Exercise Capacity in Patients With Iron Deficiency and Chronic Heart Failure                                                                                             |
| <b>FAIR-HF</b>    | Ferinject Assessment in Patients With Iron Deficiency and Chronic Heart Failure                                                                                                                             |
| <b>FBXL5</b>      | F-box and leucine-rich repeat protein 5                                                                                                                                                                     |
| <b>FCM</b>        | ferric carboxymaltose                                                                                                                                                                                       |
| <b>FERRIC-HF</b>  | Ferric Iron in Heart Failure                                                                                                                                                                                |
| <b>GPX4</b>       | glutathione peroxidase 4                                                                                                                                                                                    |
| <b>HEART-FID</b>  | Treatment for Heart Failure With Iron Deficiency                                                                                                                                                            |
| <b>IRON-5 HF</b>  | Iron Supplementation in Heart Failure Patients With Anemia                                                                                                                                                  |
| <b>IRONMAN</b>    | Effectiveness of Intravenous Iron Treatment Versus Standard Care in Patients With Heart Failure and Iron Deficiency                                                                                         |
| <b>IRONOUT</b>    | Oral Iron Repletion Effects on Oxygen Uptake in Heart Failure                                                                                                                                               |
| <b>IRP</b>        | iron regulatory protein                                                                                                                                                                                     |
| <b>MMP</b>        | matrix metalloproteinase                                                                                                                                                                                    |
| <b>NTBI</b>       | non-transferrin-bound iron                                                                                                                                                                                  |
| <b>PIVOTAL</b>    | Proactive Intravenous Iron Therapy in Hemodialysis Patients                                                                                                                                                 |
| <b>ROS</b>        | reactive oxygen species                                                                                                                                                                                     |
| <b>TACT</b>       | Trial to Assess Chelation Therapy                                                                                                                                                                           |
| <b>TFR1</b>       | transferrin receptor 1                                                                                                                                                                                      |
| <b>VSMC</b>       | vascular smooth muscle cell                                                                                                                                                                                 |

regulated by IRP1 and IRP2, which modulate the abundance of iron transporters and storage proteins, including TFR1 (transferrin receptor 1), divalent metal transporter 1 (DMT1), ferroportin-1 (FPN1), and ferritin.<sup>3</sup> Mechanistically, IRPs bind to the stem-loop structure of the iron-responsive element located in the 3' untranslated region of *TFR1* and *DMT1* mRNAs in conditions of low cellular iron levels.<sup>4</sup> This binding stabilizes *TFR1* and *DMT1* mRNAs and enhances iron absorption. In contrast, when the iron levels are elevated, IRP1 acquires a [4Fe-4S] cluster, which converts the protein into a cytosolic aconitase and loses its RNA-binding capacity.<sup>5</sup> In parallel, in iron-replete conditions, IRP2 undergoes iron- and oxygen-dependent ubiquitination and proteasomal degradation by E3 ubiquitin ligase F-box and leucine-rich repeat protein 5 (FBXL5).<sup>6</sup> Reduced IRP activity under high intracellular iron results in destabilization of *TFR1* and *DMT1* mRNAs, suppresses iron uptake, and prevents excessive intracellular iron accumulation.<sup>4</sup> The IRP/iron-responsive element system likewise limits translation of ferritin and FPN1 via iron-responsive elements located in their 5' untranslated regions in low iron conditions, thereby restricting storage and export.<sup>4</sup> The mechanisms by which IRP1 and IRP2 regulate iron metabolism-related genes are illustrated in Figure 1. Animal models support the critical role of IRPs in maintaining cellular and systemic iron homeostasis. *Irp1*<sup>-/-</sup> mice exhibit a marked transient polycythemia, characterized by reduced *HIF-2α* mRNA, increased erythropoietin (*Epo*) mRNA in kidneys, and inappropriately elevated serum erythropoietin levels. *Irp1* deletion results in extramedullary hematopoiesis, splenomegaly, polycythemia, reticulocytosis, increased hepcidin production, pulmonary hypertension, and premature death.<sup>7,8</sup> In contrast, *Irp2*-deficient mice display elevated ferritin levels, reduced TFR1 expression, impaired iron uptake, and progressive adult-onset neurodegeneration.<sup>9</sup> Overall, while IRP1 and IRP2 share overlapping roles in regulating iron metabolism, they work through different regulatory mechanisms, and the loss of either of them exhibits distinct physiological outcomes. In addition, as a critical ferroxidase, ceruloplasmin catalyzes iron oxidation from ferrous iron (Fe<sup>2+</sup>) to ferric iron (Fe<sup>3+</sup>).<sup>10</sup> This process enables iron binding onto transferrin and facilitates cellular iron efflux via ferroportin-1. Ceruloplasmin dysfunction impairs iron export and leads to tissue iron overload and oxidative injury, which contribute to neurodegenerative conditions such as Parkinson disease.<sup>11</sup> In mice, ceruloplasmin knockout induced iron accumulation in multiple tissues and organs (liver, spleen, pancreas, and duodenum), indicating its essential role in systemic iron homeostasis.<sup>12</sup>

(IRPs), specifically IRP1 (encoded by *ACO1*) and IRP2 (encoded by *IREB2*), represent the critical intracellular iron sensors that directly regulate iron metabolism-related genes.<sup>3</sup> Cellular iron homeostasis is meticulously

## Iron Overload

Iron overload is the condition of excessive accumulation of iron in various organs resulting from increased



**Figure 1.** IRP1- and IRP2-mediated regulation of iron metabolism-related genes.

In iron-deficient cells, IRP1 and IRP2 bind to IREs in the 3' untranslated regions of TFR1 and DMT1 mRNAs, stabilizing these transcripts and promoting iron uptake. When iron levels are sufficient, IRP1 incorporates a [4Fe-4S] cluster and functions as cytosolic aconitase, while IRP2 undergoes iron- and oxygen-dependent degradation via FBXL5, resulting in loss of IRE binding. Reduced IRP activity leads to destabilization of TFR1 and DMT1 mRNAs, preventing excess iron accumulation. IRP binding also suppresses translation of FPN1 and ferritin under low-iron conditions, limiting iron export and storage. Collectively, the IRP-IRE system ensures precise regulation of iron import, export, and sequestration to maintain cellular iron homeostasis. DMT1 indicates divalent metal transporter 1; FBXL5, F-box and leucine-rich repeat protein 5; FPN1, ferroportin-1; IRE, iron-responsive element; IRP, iron regulatory protein; and TFR1, transferrin receptor 1.

intestinal absorption, parenteral administration (eg, blood transfusions), or excessive dietary consumption.<sup>13</sup> It is primarily caused by a genetic defect in iron homeostasis, referred to as hereditary hemochromatosis (HH), that can lead to liver cirrhosis, diabetes, arthritis, endocrine dysfunction, and cardiomyopathy.<sup>14</sup> Clinical indicators of iron overload include elevated serum ferritin levels, increased serum iron, and high transferrin saturation.<sup>15</sup> In iron overload conditions, iron levels in the circulation exceed the iron-binding capacity of serum transferrin, resulting in non-transferrin-bound iron (NTBI) uptake. Iron overload plays a positive role in the development of atherosclerosis.<sup>16</sup> It negatively impacts cardiovascular health in patients with hemochromatosis, exhibiting increased morbidity and death.<sup>17</sup> Table 1 summarizes<sup>18–22</sup> the commonly used screening and diagnostic criteria for iron overload, particularly in the context of hereditary hemochromatosis (American Association for the Study of Liver Diseases 2011 Practice Guideline, Diagnosis and Management

of Hemochromatosis).<sup>18</sup> Criteria include elevated transferrin saturation<sup>19,20</sup> and serum ferritin levels,<sup>21,22</sup> followed by confirmatory testing to rule out secondary causes and assess tissue iron stores. In addition to the conventional measures above, emerging biomarkers of iron dysregulation, including labile plasma iron,<sup>23</sup> NTBI,<sup>24</sup> and ferritin isoforms (H- and L-ferritin),<sup>25</sup> can provide additional insights into oxidative stress and vascular injury.

## Iron Deficiency

Iron deficiency commonly arises from inadequate dietary intake, impaired intestinal absorption, increased iron loss, or enhanced physiological demand. It negatively affects physical function, cognitive performance, and the quality of life.<sup>26</sup> Iron deficiency is prevalent in patients with CVD, including chronic heart failure (HF), acute HF, pulmonary arterial hypertension, chronic obstructive pulmonary disease, abdominal aortic aneurysm (AAA), and aortic stenosis.<sup>26,27</sup> The prevalence of the

**Table 1. Screening and Diagnostic Criteria for Iron Overload<sup>18</sup>**

| Terms of iron overload                          | Test                                                                                                                                                                                                                                                                    | Men           | Women         |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|
| Screening-positive for elevated iron measures   | TSAT <sup>19,20</sup>                                                                                                                                                                                                                                                   | >50%          | >45%          |
|                                                 | Serum ferritin level <sup>21,22</sup>                                                                                                                                                                                                                                   | >300 µg/L     | >200 µg/L     |
| Possible iron overload                          | Repeat TSAT and serum ferritin<br>Initial increased TSAT and serum ferritin levels plus clinical examination                                                                                                                                                            | Same as above | Same as above |
| Provisional primary iron overload <sup>20</sup> | Repeated TSAT and increased serum ferritin levels with exclusion of secondary causes of iron overload (eg, liver disease, inflammation, alcohol use)                                                                                                                    |               |               |
| Documented iron overload <sup>20</sup>          | Meets all the PPIO criteria plus ≥1 of the following criteria:<br>Hepatic iron concentration (from biopsy) ≥90 µm/g (≥5000 µg/g dry weight)<br>Iron removed by phlebotomy >4 g<br>Histology examination:<br>Hepatic iron index ≥1.9<br>Hepatic iron staining: 3+, or 4+ |               |               |

PPIO indicates provisional primary iron overload; and TSAT, transferrin saturation.

condition is more frequently observed in women.<sup>27</sup> To evaluate the therapeutic potential of iron supplements, it is necessary first to see the impact of iron deficiency on cardiovascular function. Recent research has shed light on the roles of hepcidin and IRPs in maintaining iron homeostasis in this context.<sup>26</sup> Hepcidin production is reduced in iron deficiency, hemorrhagic/hemolytic anemia, and hypoxic conditions, enhancing intestinal iron absorption and increasing iron release from the body's iron stores. However, in case of iron overload and an inflammatory state, hepcidin is upregulated in response to inflammatory cytokines, particularly interleukin-1β, interleukin-6, and interleukin-22.<sup>28</sup> Increased hepcidin leads to iron sequestration in macrophages and hepatocytes and reduces iron exports from enterocytes. At the posttranscriptional level, the iron-saturated IRPs lose their capacity to bind to the 5' untranslated region of ferritin mRNA, resulting in decreased ferritin abundance and impaired iron storage.<sup>29</sup> Table 2 outlines the stages of iron deficiency in humans, key diagnostic indicators, and their

respective reference ranges (laboratory measurements commonly used in the evaluation of iron status (Institute of Medicine [US] Panel on Micronutrients, 2001).<sup>30,31</sup> These stages progress from early iron depletion to iron deficiency anemia, identified by distinct biochemical and hematologic markers.

## IRON DYSREGULATION IN VASCULAR DISEASES

Given the crucial role of iron in various enzymes and physiological functions, dysregulation of iron metabolism (deficiency and overload) is a pathophysiologic mechanism for a wide range of diseases.<sup>2</sup> In contrast with other forms of classical unprogrammed cell death, ferroptosis is predominantly associated with disruptions in iron metabolism, lipid peroxidation, and imbalances in the antioxidant system, as discussed in other reviews.<sup>32</sup> Recent studies have demonstrated the role of ferroptosis in the pathogenesis of various CVDs.<sup>32</sup>

**Table 2. Iron Deficiency Stages**

| Stages of iron deficiency                                       | Test/indicator                  | Diagnostic range      |
|-----------------------------------------------------------------|---------------------------------|-----------------------|
| Iron storage depletion                                          | Stainable bone marrow iron      | Absent                |
|                                                                 | Total iron-binding capacity     | >400 µg/dL            |
|                                                                 | Serum ferritin                  | <12 µg/L              |
| Iron-restricted erythropoiesis/early functional iron deficiency | Transferrin saturation          | <16%                  |
|                                                                 | Free erythrocyte protoporphyrin | >70 µg/dL erythrocyte |
|                                                                 | Soluble transferrin receptor    | >8.5 mg/L             |
|                                                                 | Total iron-binding capacity     | >360 µg/dL            |
| Iron deficiency anemia                                          | Hemoglobin (men)                | <12–14 g/100 mL       |
|                                                                 | Hemoglobin (women)              | <12 g/100 mL          |
|                                                                 | Mean cell volume                | <80 fL                |

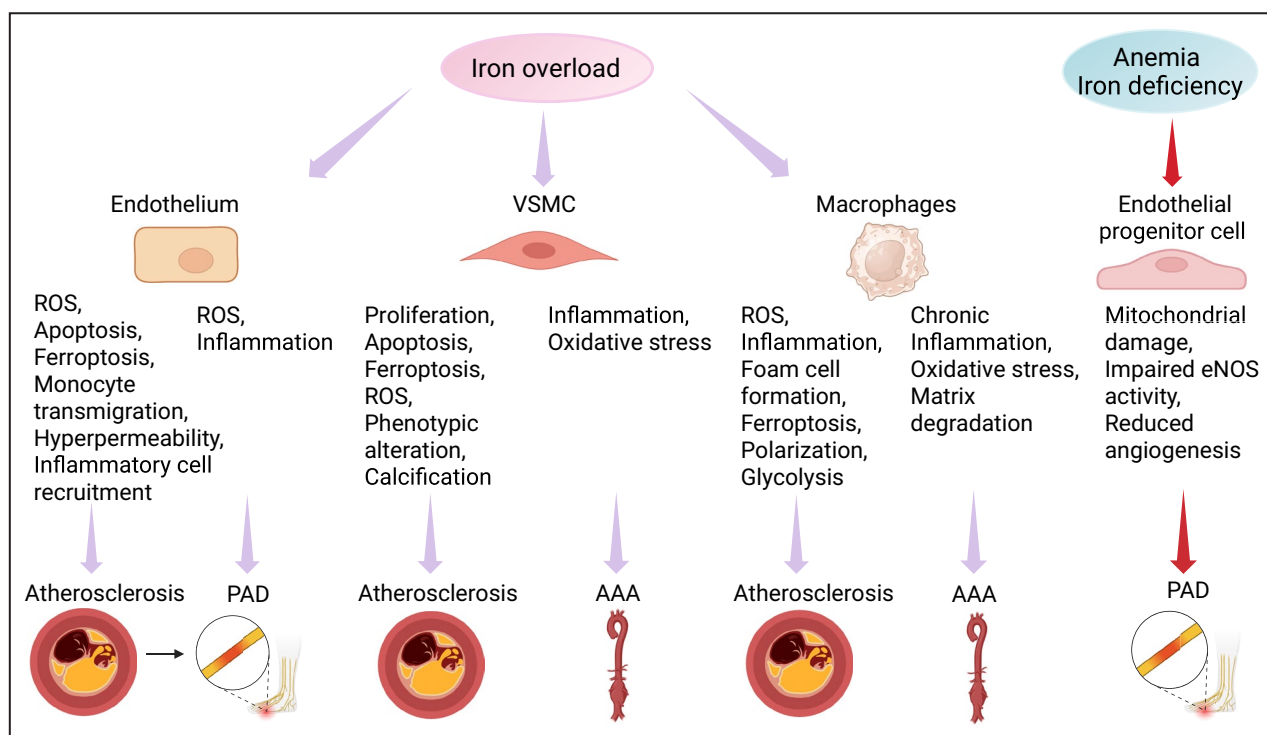
Here, we review the literature to elucidate the complex relationship between iron dysregulation and vascular diseases, including atherosclerosis, aneurysm, and PAD (Figure 2).

## Iron Dysregulation in Endothelial Cells

Endothelial cell (EC) dysfunction caused by hemodynamic changes or detrimental stimuli has been linked to atherosclerosis, coronary artery disease, PAD, and aortic aneurysm. Iron overload leads to endothelial dysfunction through pro-oxidant and proinflammatory effects, increasing monocyte adhesion, transmigration, and apoptosis, and increasing hyperpermeability, while reducing nitric oxide bioavailability.<sup>33</sup> Recent investigations have elucidated the influence of iron dysregulation on EC function and atherosclerosis.<sup>2</sup> When systemic iron levels exceed the transport capability of transferrin, NTBI circulates throughout the body and deposits in vascular cells, potentially causing endothelial oxidative damage. Epidemiological and experimental studies have demonstrated that iron overload in vascular cells promotes atherosclerosis by inducing oxidative stress, EC apoptosis, and inflammation.<sup>1</sup> Notably, iron chelation can mitigate the endothelial dysfunction through suppression of inflammatory response by inhibition of intercellular adhesion molecule 1, vascular cell adhesion

molecule 1, E-selectin, and monocyte chemoattractant protein 1 (MCP-1), suggesting that iron-induced endothelial dysfunction is reversible.<sup>34</sup> Deferoxamine, an iron chelator, improves endothelium-dependent, nitric oxide-mediated vasodilation in patients with coronary artery disease.<sup>35</sup>

Ferroptosis has also been implicated in the pathogenesis of atherosclerosis. A recent report suggests that inhibition of ferroptosis reduces endothelial dysfunction and lipid peroxidation in mouse aortic ECs and alleviates atherosclerosis in mice.<sup>36</sup> In *Apoe*<sup>-/-</sup> mice, glutathione peroxidase 4 (GPX4) overexpression reduces the development of atherosclerosis and delays its progression. Overexpression of GPX4 inhibits the expression of adhesion molecules in ECs, and the adhesion of monocytes to ECs, as well as endothelial necrosis and apoptosis.<sup>36</sup> Additionally, blocking ferroptosis by the inhibitor ferrostatin-1 prevents lipid peroxidation and reduces atherosclerosis in the aorta of *Apoe*<sup>-/-</sup> mice fed a high-fat diet. In vitro studies suggest that blocking ferroptosis inhibits oxidized low-density lipoprotein (LDL)-induced lipid peroxidation and endothelial dysfunction in mouse aortic ECs.<sup>37</sup> Another recent study reported that overexpression of sestrin 1, a stress-inducible protein that protects cells against oxidative stress and other stressors, attenuates atherosclerosis in the *Apoe*<sup>-/-</sup> mouse model



**Figure 2. Iron dysregulation contributes to the development of atherosclerosis, aortic aneurysm, and PAD.**

Both iron overload and iron deficiency impair the function of ECs, VSMCs, macrophages, and endothelial progenitor cells, leading to the pathogenesis of vascular diseases. The underlying mechanisms reveal how iron imbalance drives these vascular disorders. AAA indicates abdominal aortic aneurysm; EC, endothelial cell; eNOS, endothelial nitric oxide synthase; PAD, peripheral artery disease; ROS, reactive oxygen species; and VSMC, vascular smooth muscle cell.



and suppresses inflammation, oxidative stress, and ferroptosis in human umbilical vein ECs treated with oxidized LDL.<sup>38</sup> A recent study revealed that 59 genes associated with ferroptosis were identified in human atherosclerotic plaques by reanalysis of Gene Expression Omnibus data sets. Through functional annotation, including Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment, and construction of protein–protein interaction networks, 5 hub genes (mitogen-activated protein kinase 1[MAPK1], tumor protein p53[TP53], signal transducer and activator of transcription 3[STAT3], prostaglandin-endoperoxide synthase 2[PTGS2], and heme oxygenase 1[HMOX1]) were identified.<sup>39</sup> Nevertheless, the role of ferroptosis in atherosclerosis and specific molecular mechanisms remains to be fully explored. The role of endothelial iron dysregulation in atherosclerosis is summarized in Figure 2. Sex hormones profoundly affect cellular and systemic iron balance and impact vascular disease. Premenopausal women typically have lower iron stores due to menstrual blood loss and estrogen-mediated suppression of hepatic hepcidin. In contrast, men and postmenopausal women tend to accumulate higher tissue iron, partially through testosterone-driven hepcidin down-regulation and increased erythropoiesis.<sup>40</sup> Serum ferritin levels are positively and independently associated with carotid atherosclerosis in Asian women but not in Asian men.<sup>41</sup> Additionally, intravenous saccharated ferric oxide significantly increases iron and malondialdehyde, a marker of lipid oxidation, and impairs flow-mediated vasodilatation, an index for EC function, in male nonsmokers.<sup>42</sup>

## Iron Dysregulation in Vascular Smooth Muscle Cells

Dysfunction of vascular smooth muscle cells (VSMCs), a major cell type in the vessel tunica media, contributes to atherosclerotic plaque formation and aortic aneurysm development. VSMCs exhibit remarkable phenotypic plasticity, readily phenotypically switching to macrophage-like or mesenchymal stem cell-like cells, which contributes to vascular remodeling and disease progression.<sup>43</sup> Iron overload induces malfunction in VSMCs, including proliferation, apoptosis, ROS production, phenotypic switch, and calcification.<sup>44</sup>

Iron overload leads to extensive accumulation of iron in the arterial media layer (rich in VSMCs) in *Apoe*<sup>−/−</sup>/*Fpn1* wt/C326S mice, a genetically engineered double-mutation model combining *Apoe* deficiency with *Fpn1* gain of function. The collagen content in atherosclerotic plaques in *Apoe*<sup>−/−</sup>/*Fpn1* wt/C326S mice was reduced, even though there was no significant change in the quantity of VSMCs in the plaques.<sup>16</sup> Accumulation of NTBI in VSMCs creates a pro-oxidant

environment, promoting foam cell formation and increasing plaque vulnerability. Treatment with ferric ammonium citrate, an iron supplement delivering a bioavailable source of ferric iron, induces apoptosis, inflammation, and MCP-1 expression, as well as monocyte adhesion and migration.<sup>16</sup> Moreover, iron chelation using deferoxamine significantly reduces VSMC proliferation.<sup>45</sup> Although elevated ferritin levels, reflecting iron dysregulation, are positively associated with cardiovascular risk in patients with chronic kidney disease,<sup>46</sup> clinical studies that directly link iron overload to vascular calcification remain limited. Cell culture studies indicate that iron-induced calcification in human aortic smooth muscle cells is mediated by interleukin-24, which is increased during calcification.<sup>47</sup> Consistently, in vivo studies demonstrated that iron overload promotes aortic calcium deposition and interleukin-24 expression in rats, which further supports interleukin-24's role in pathogenic vascular calcification.<sup>48</sup> Moreover, both solute carrier family 39 member 14 and solute carrier family 39 member 8 are upregulated in VSMCs during vascular calcification. Overexpression of solute carrier family 39 member 14 enhances calcification and increases intracellular iron accumulation in VSMCs.<sup>49</sup> *Zip14* is also upregulated in macrophages in response to lipopolysaccharide treatment. Silencing of *Zip14* in macrophages significantly reduced the production and release of proinflammatory cytokines, including interleukin-6 and tumor necrosis factor- $\alpha$  (TNF- $\alpha$ ).<sup>50</sup> Taken together, solute carrier family 39 member 14 promotes iron and other metal uptake, enhances vascular calcification, and induces inflammation in diverse cell types. However, the precise mechanisms by which iron contributes to vascular calcification remain to be fully elucidated.

Iron-induced inflammation and oxidative stress contribute to the pathophysiology of AAA. The iron-accumulated area positively correlates with the macrophage infiltration area in both human and murine AAA walls. Iron restriction decreases the incidence of AAA with a reduction of oxidative stress and inflammation.<sup>51</sup> Smooth muscle cell-specific deletion of *Smad4* increases the expression of inflammatory cytokine interleukin-1 $\beta$ , which promotes vascular inflammation and AAA formation in adult mice.<sup>52</sup> Recent work by Ren et al. revealed the critical role of ferroptosis in the formation and rupture of AAA, and the association of ferroptosis with inflammatory and oxidative responses through TNF and the NOD-like signaling pathways. GPX4 was identified as a critical anti-ferroptosis gene and a potential therapeutic target for AAA treatment.<sup>53</sup> Supporting this, recent investigations in *Apoe*<sup>−/−</sup> mice found that activation of the solute carrier family 7 member 11–GPX4 signaling pathway by ferrostatin-1 inhibits lipid peroxidation and ferroptosis in mouse VSMCs and improves structural integrity of the aortic wall.<sup>54</sup> The

mechanisms by which iron dysregulation in VSMCs contributes to atherosclerosis and AAA are illustrated in Figure 2. Currently, few clinical studies have confirmed the effects of ferroptosis on AAA progression, underscoring the opportunity and need to clarify its therapeutic potential.

## Iron Dysregulation in Macrophages

Macrophages are involved in early atherosclerotic lesion formation and advanced plaque stability by inflammatory and lipid-handling pathways. Iron dysregulation, both iron deficiency and iron overload, induces ROS and lipid peroxidation in macrophages, thereby exacerbating vascular inflammation. Iron overload induces macrophage dysfunction by promoting foam cell formation and, as recently reported, by altering glycolysis and macrophage polarization.<sup>55</sup>

Notably, the iron chelator deferoxamine decreases intracellular iron levels in *Apoe*<sup>-/-</sup> mice and reduces proinflammatory factors, including MCP-1, and markers of macrophage infiltration. This intervention effectively attenuates atherosclerotic lesion formation.<sup>56</sup> Similarly, treatment with deferiprone, another iron chelator, reduced lipopolysaccharide-induced M1 phenotypic expression and decreased oxidized LDL uptake.<sup>57</sup> Hepcidin deficiency is associated with increased serum iron and decreased intracellular iron in macrophages. Hepcidin- and LDL receptor-deficient (*Hamp*<sup>-/-</sup>/*Ldlr*<sup>-/-</sup>) mice fed a high-fat diet for 21 weeks showed attenuated atherosclerosis and reduced iron content and inflammatory response in macrophages.<sup>57</sup> Macrophage-specific *Fpn1* deficiency (*Fpn1*<sup>LysM</sup>/*LysM*/*Apoe*<sup>-/-</sup>) was shown to exacerbate atherosclerosis by increasing ROS and systemic inflammation and altering plaque lipid composition. Mechanistically, *Fpn1* deficiency significantly downregulates ABC transporters (ABCA1 and ABCG1) via reducing the expression of liver X receptor- $\alpha$ .<sup>58</sup> In contrast, another study using *Apoe*<sup>-/-</sup> mice with macrophage-specific *Fpn1* deletion showed iron accumulation in macrophages without a significant effect on overall atherosclerotic lesion development.<sup>59</sup> A separate investigation using *Apoe*<sup>-/-</sup>/*Fpn1* wt/C326S mice demonstrated increased macrophage content and lipid burden within atherosclerotic plaques. Additionally, this study suggests that intraplaque VSMCs may adopt a macrophage-like phenotype through a phenotypic switch under conditions of iron dysregulation.<sup>16</sup> Collectively, these findings indicate that the relationship between macrophage iron metabolism and atherosclerosis progression is still incompletely understood and needs more studies.<sup>16,33</sup> Single-nucleotide polymorphisms in the hemochromatosis gene *Hfe* (homeostatic iron regulator) have been identified to be associated with reduced LDL cholesterol in human plasma on the basis of analysis

of the Global Lipids Genetics Consortium dataset. In *Apoe*<sup>-/-</sup> mice, *Hfe* deletion reduced atherosclerosis. Mechanistically, *Hfe* not only acts as a negative regulator of LDL receptor expression in hepatocytes but also controls the Kupffer cell uptake and transfer of LDL cholesterol to hepatocytes.<sup>60</sup> The study links iron availability, Kupffer cell function, cholesterol homeostasis, and atherosclerosis.

Macrophages mediate iron accumulation in AAA tissues, and chronic inflammation, oxidative stress, and extracellular matrix degradation contribute to disease progression.<sup>51,61</sup> Recalcati et al. demonstrated that iron-loaded macrophages upregulate matrix metalloproteinases (MMPs) under proinflammatory conditions.<sup>62</sup> Inhibiting macrophage accumulation and reducing inflammation may protect against AAA development.<sup>61</sup> Kitagawa et al. revealed that AAA formation in *Apoe*<sup>-/-</sup> mice induced by angiotensin II positively correlates with iron accumulation in both vascular macrophages and ECs, a feature absent in control mice.<sup>63</sup> Similarly, Sawada et al. demonstrated that iron accumulation and macrophage infiltration occur in human and murine AAA tissues. Iron restriction decreases the incidence of AAA formation by suppressing inflammation and oxidative stress. Aortic expression of *TFR1* was increased in human and murine AAA walls, and the TFR1-positive area correlates with regions of iron accumulation and F4/80-positive macrophage infiltration.<sup>51</sup> Moreover, iron restriction attenuates the activities of c-Jun N-terminal kinase, aortic MMP-2, and MMP-9 in mouse AAA walls.<sup>51</sup> Single-cell RNA sequencing analysis of human AAA tissues revealed significant upregulation of the ferroptosis-related pathway in macrophages compared with normal tissue. In murine AAA models, treatment with liproxstatin-1, a specific ferroptosis inhibitor, reduces aortic diameter, suppresses proinflammatory cytokine expression, and limits macrophage infiltration. At the mechanistic level, liproxstatin-1 mitigates ferroptosis and lowers macrophage MMP-9 expression.<sup>64</sup> How macrophage iron dysfunction contributes to the pathogenesis of atherosclerosis and AAA is illustrated in Figure 2. In summary, correction of iron dysregulation in macrophages is a potential therapeutic approach to slow AAA progression.

## Iron Dysregulation in Other Immune Cell Types

In addition to macrophages, iron metabolism also modulates the activity of other immune cells that contribute to vascular injury. In neutrophils, iron is crucial for antimicrobial function and for the formation of neutrophil extracellular traps (NETs).<sup>65</sup> Neutrophils significantly contribute to atherosclerosis by secreting interleukin-1 $\beta$ . Using the murine model of atherosclerosis, cholesterol crystals triggered neutrophils to release

neutrophil extracellular traps. These neutrophil extracellular traps prepare macrophages to release cytokines, activate T-helper 17 cells, and promote sterile inflammation in atherosclerosis.<sup>66</sup> The research indicated that hyperlipidemia in the *Apoe*-deficient mouse model leads to neutrophilia, which is associated with the early development of atherosclerotic lesions. Neutrophils are also present at sites of plaque rupture and in individuals experiencing acute coronary syndromes.<sup>67</sup> Iron availability is essential for T-cell activation and effector function. Iron influx supports interleukin-2 receptor signaling, mitochondrial respiration, and proliferation of CD4<sup>+</sup> T cells. Consistently, iron restriction during T-cell stimulation negatively impacts mitochondrial function.<sup>68</sup> The role of ferroptosis in T cells has been explored primarily in the field of tumor immunology. Regulation of ferroptosis in T cells, including how iron availability, lipid peroxidation, and GPX4 influence T-cell fate and function, remains incompletely understood.<sup>69</sup> Direct evidence linking iron dysregulation in T cells to vascular diseases is currently limited. Natural killer cells have been implicated in a range of vascular diseases, mainly through their effects on immune surveillance and vascular inflammation. Current studies show that iron availability modulates natural killer metabolism and cytotoxicity, mainly in viral infection models.<sup>70</sup> Direct evidence that natural killer cell-specific iron dysregulation causes vascular disease remains limited. Collectively, accumulating data suggest that iron dysregulation in immune cells may promote proinflammatory polarization and potentially facilitate vascular pathology, which warrants further investigation.

## SYSTEMIC IRON DYSREGULATION AND VASCULAR DISEASES

The direct effects of iron dysfunction on vascular cells, associated with atherosclerosis and aortic aneurysms, have been increasingly well recognized; however, its effects on vascular cells in PAD are less well understood. PAD is a progressive vascular disease characterized by impaired blood flow and ischemia, primarily caused by atherosclerosis.<sup>71</sup> Indeed, systemic iron overload or deficiency is implicated in the progression of PAD.

Iron overload promotes oxidative stress and vascular inflammation. Excess reactive iron facilitates ROS formation via Fenton chemistry, resulting in endothelial dysfunction, lipid peroxidation, and VSMC dysfunction.<sup>72</sup> Elevated serum ferritin has been associated with an increased risk of atherosclerotic disease, including PAD.<sup>73</sup> NTBI promotes oxidative stress and disrupts cellular membrane lipids, thereby impairing cell function. In patients with PAD, the NTBI pool shifts from low-molecular-weight complexes to higher-molecular-weight forms compared with healthy controls. Most

NTBI is bound to serum albumin, and this NTBI-albumin complex can be isolated and may serve as a potential biomarker.<sup>74</sup> Sex differences in PAD have been well recognized. Women tend to present later in the disease course with greater functional impairment and more rapid decline than men. Although men generally have higher rates of lower-limb amputation, women often experience poorer recovery and quality of life.<sup>75</sup> Notably, a recent study found that increased ferritin within physiological limits is associated with clinically significant PAD in women, suggesting a possible role of iron (directly or indirectly) and highlighting the sex differences in PAD.<sup>76</sup> Iron chelation or reduction of iron stores can attenuate vascular lesions and improve endothelial function.<sup>35</sup> Conversely, iron deficiency, often observed in chronic inflammation and kidney disease, is associated with impaired EC angiogenesis and proliferation.<sup>77</sup> Iron is required to maintain mitochondrial energy production and endothelial nitric oxide synthase activity.<sup>78</sup> Iron deficiency can worsen ischemia in PAD by inhibiting endothelial function recovery and neovascularization.<sup>79</sup> In the mouse ischemic hind-limb model, deferoxamine pretreatment accelerates endothelial progenitor cell senescence, suppresses endothelial progenitor cell angiogenesis, and inhibits blood perfusion in ischemic tissue.<sup>80</sup> The impact of endothelial iron dysfunction on PAD is depicted in Figure 2. Recent evidence shows increased expression of ferroptosis-related genes, including prostaglandin-endoperoxide synthase 2, heme oxygenase 1, acyl-CoA synthetase long-chain family member 4, ELAV-like RNA-binding protein 1, and beclin-1, in the calf muscle of patients with critical limb ischemia PAD compared with controls without PAD and patients with intermittent claudication, implying that ferroptosis contributes to myopathy at an advanced stage of PAD.<sup>81</sup> Overall, these findings underscore the importance of precise iron regulation to support effective vascular repair in ischemic conditions and highlight the need to dissect the mechanisms by which iron dysregulation in vascular cells drives PAD pathogenesis.

Anemia is a common condition in patients hospitalized for PAD and is independently associated with adverse clinical outcomes, including higher rates of major adverse cardiovascular and limb events, amputation, and death. Both amputation and mortality risk increase significantly at low levels of hemoglobin.<sup>82</sup> Iron deficiency anemia is highly prevalent worldwide. Iron deficiency, particularly transferrin saturation levels, may also exacerbate the severity of PAD, and correcting iron deficiency may improve outcomes, suggesting that iron deficiency could be a therapeutic target.<sup>83,84</sup> Anemia increases the risk of perioperative morbidity and death among patients with PAD, who have an increased prevalence of iron deficiency anemia. Patients with critical ischemia, the most severe category of



symptomatic PAD, also have a higher prevalence of anemia, as well as high near-term rates of amputation and death.<sup>84,85</sup> More research is needed to fully understand the relationship among anemia, iron deficiency, and PAD, which could help determine the best strategies for managing iron deficiency in PAD.<sup>85</sup>

## IRON STATUS IN ACUTE CEREBRAL DISEASES AND CVDS

Iron balance is increasingly recognized as a determinant of injury severity and recovery in acute stroke and myocardial ischemia. A Mendelian randomization analysis showed that genetically higher serum iron, ferritin, and transferrin saturation are positively associated with stroke.<sup>86</sup> In contrast, iron deficiency has emerged as a prognostic factor in acute cardiovascular events. A meta-analysis of acute coronary syndrome cohorts found that iron deficiency is independently associated with a worse long-term prognosis.<sup>87</sup> Furthermore, a recent prospective multicenter cohort demonstrated that low ferritin at admission, defined as <100 or 100 to 299 ng/mL with transferrin saturation <20%, was associated with worse 90-day functional recovery in acute ischemic stroke.<sup>88</sup> These findings support that both iron overload and deficiency impair tissue resilience, underscoring the importance of maintaining iron homeostasis in acute cerebrovascular disease and CVD.

## POTENTIAL IRON THERAPIES AND CLINICAL TRIALS

### Therapeutic Agents Targeting Iron-Related Pathways in Atherosclerosis

Growing evidence suggests that therapies targeting iron metabolism may offer a promising strategy for treating atherosclerosis and other CVDs. Excess iron triggers multiple cellular disturbances, including oxidative stress, inflammation, and ferroptosis, which in turn contribute to endothelial dysfunction, VSMC calcification, and foam cell formation. Deferoxamine, deferiprone, deferasirox, and dextrazoxane are Food and Drug Administration–approved iron chelators widely used in clinical practice to manage iron overload disorders.<sup>89</sup> In parallel, ferroptosis inhibitors, including ferrostatin-1 and liproxstatin-1, have emerged as potential agents to prevent the progression of atherosclerosis. Those inhibitors scavenge lipid peroxides, reduce labile iron pools, and limit ROS accumulation, thereby targeting the iron-dependent lipid peroxidation pathways.<sup>90</sup> Beyond traditional iron chelation and supplementation, several new strategies can potentially modulate iron handling more precisely in vascular disease. Modifying

the hepcidin–ferroportin-1 axis with hepcidin agonists or antagonists would redistribute iron from macrophages and vascular cells, thereby reducing inflammation and inhibiting atherogenesis.<sup>57</sup> Radical-trapping antioxidants ferrostatin-1 and liproxstatin-1, both ferroptosis inhibitors, reduce iron-induced lipid peroxidation and attenuate vascular injury in preclinical models of atherosclerosis and ischemia–reperfusion injury.<sup>37,91</sup> Emerging approaches that target ferritinophagy, lysosomal iron regulation, and mitochondrial iron handling further highlight iron metabolism as a viable therapeutic pathway to mitigate vascular injury in atherosclerosis, aneurysm, and PAD.

### Clinical Trials Evaluating Oral and Intravenous Iron Therapy

Clinical trials evaluating iron therapy in CVD, particularly HF, have been conducted, and some have shown beneficial outcomes for patients. Several studies have demonstrated the benefits of intravenous iron therapy in HF, particularly in improving exercise and functional capacity. The FERRIC-HF (Ferric Iron in Heart Failure) trial (n=35) showed improved exercise tolerance, especially in anemic patients receiving intravenous iron.<sup>92</sup> The FAIR-HF (Ferinject Assessment in Patients With Iron Deficiency and Chronic Heart Failure) trial was the very first randomized controlled trial to show a functional benefit of intravenous iron in patients with HF, conducted by a large double-blind, placebo-controlled study, randomized 459 patients with HF with reduced ejection fraction and iron deficiency to receive ferric carboxymaltose (FCM) or placebo, showing improved symptoms and exercise capacity.<sup>93</sup> CONFIRM-HF (A Study to Determine the Efficacy and Safety of Finerenone and SGLT2i in Combination in Hospitalized Patients With Heart Failure; n=304) further demonstrated that FCM significantly improved 6-minute walk distance, reduced hospitalizations, and enhanced quality of life.<sup>94</sup> In EFFECT-HF (Effect of Ferric Carboxymaltose on Exercise Capacity in Patients With Iron Deficiency and Chronic Heart Failure; n=172), FCM maintained peak VO<sub>2</sub> over 24 weeks, but the control group declined.<sup>95</sup> The AFFIRM-AHF (A Randomized, Double-Blind Placebo Controlled Trial Comparing the Effect of Intravenous Ferric Carboxymaltose on Hospitalizations and Mortality in Iron Deficient Patients Admitted for Acute Heart Failure) trial evaluated intravenous FCM in patients with iron deficiency hospitalized for acute HF. While FCM reduced recurrent hospitalizations, it did not significantly lower cardiovascular death over 1 year.<sup>96</sup> In the IRONMAN (Effectiveness of Intravenous Iron Treatment Versus Standard Care in Patients With Heart Failure and Iron Deficiency) trial (2022), 1137 patients with chronic HF and iron deficiency were followed up for up to 7 years. Although intravenous iron

was associated with fewer HF hospitalizations and cardiovascular deaths, the differences were not statistically significant.<sup>97</sup> The PIVOTAL (Proactive Intravenous Iron Therapy in Hemodialysis Patients) trial, conducted in hemodialysis patients, showed that high-dose intravenous iron significantly reduced the risk of first and recurrent HF events.<sup>98</sup> In this population, these clinical benefits may result from correction of functional iron deficiency and enhancement of systemic oxygen delivery, which could outweigh potential proatherogenic effects of iron supplementation. The impact of iron supplementation likely varies with the clinical context, baseline iron status, and patient comorbidities. Conversely, in the TACT (Trial to Assess Chelation Therapy), disodium EDTA-based iron chelation therapy was evaluated in patients with atherosclerosis and was shown to reduce adverse cardiovascular events in stable patients following myocardial infarction.<sup>99</sup>

The HEART-FID (Treatment for Heart Failure With Iron Deficiency) trial (2023; n=3065) demonstrated that intravenous FCM modestly improved 6-minute walk distance and reduced HF-related hospitalizations and deaths over 12 months.<sup>100</sup> The recent FAIR-HF2 trial found that patients with transferrin saturation <20% showed comparable benefits regardless of transferrin saturation levels and no differential response based on ischemic status. Importantly, higher cumulative intravenous iron doses were well tolerated, addressing prior safety concerns. However, the overall results did not meet predefined statistical criteria for efficacy.<sup>101</sup> Recent clinical trials indicate that treating iron deficiency in patients with HF reduces the risk of adverse cardiovascular events, but optimal intravenous iron dosing strategies remain to be established. Intravenous iron therapy appears more effective than oral supplementation for improving functional capacity in HF. In the IRON-5 HF (Iron Supplementation in Heart Failure Patients With Anemia) trial, intravenous iron sucrose produced more favorable improvements in peak VO<sub>2</sub> than high-dose oral ferrous sulfate.<sup>102</sup> In contrast, the IRONOUT (Oral Iron Repletion Effects On Oxygen Uptake in Heart Failure) trial found that an oral polysaccharide iron complex did not lead to a significant change in peak VO<sub>2</sub> compared with placebo.<sup>103</sup> Table 3 summarizes clinical trials of iron supplementation in HF.<sup>92–103</sup>

## CURRENT CHALLENGES AND FUTURE DIRECTIONS

Iron imbalance leads to vascular dysfunction and vascular disease, with ROS and inflammatory processes being key components. Understanding the role of iron dysregulation in vascular diseases might yield efficient strategies for treating them.<sup>2</sup> Targeting iron

dysregulation has shown potential in mitigating the inflammatory aspects of cardiac disease. For example, iron chelator deferiprone effectively attenuates inflammatory responses in rats with dilated cardiomyopathy by reducing nuclear factor  $\kappa$ -light-chain-enhancer of activated B cells and cyclooxygenase-2 transcript and protein levels.<sup>104</sup> Beyond iron chelation and supplementation, additional strategies aimed at modulating systemic and cellular iron balance include targeting the prolyl hydroxylase domain protein–hypoxia-inducible factor pathway, regulating the hepcidin–ferroportin-1 axis, and using ferroptosis inhibitors.<sup>1</sup> Modulating systemic or cellular iron levels through hepcidin regulation, iron chelation, or ferroptosis inhibition may provide new avenues to inhibit oxidative vascular injury. Further studies are needed to evaluate therapeutic approaches that restore iron homeostasis to preserve vascular function. Combining such strategies with metabolic or anti-inflammatory interventions may provide synergistic protection against vascular injury.

In atherosclerosis, current investigations focus on elucidating multiple aspects of iron biology. However, the effects of various iron sources on vascular pathology, the prognostic value of NTBI and transferrin-bound iron, and the impact of ferroptosis on vascular dysfunction remain to be fully investigated.<sup>33</sup> In addition, AAA and atherosclerosis share common risk factors and pathological features. It is therefore plausible that preventing iron accumulation may warrant investigation as a therapeutic strategy for AAA. Indeed, iron restriction has been shown to suppress AAA formation by reducing inflammatory responses and oxidative stress and by improving endothelial function.<sup>51,61</sup> Moreover, sex differences in iron status have been reported in vascular diseases, including atherosclerosis and PAD. The mechanisms underlying sex-dependent regulation of iron handling and differential cellular responses remain unclear and warrant further investigation.

Integrated advanced approaches combining analyses of human samples with preclinical animal models and appropriate alternative in vitro systems will provide a robust framework to uncover novel mechanisms of iron dysregulation in vascular injury. Single-cell and single-nucleus RNA sequencing, spatial transcriptomics, and metabolomics can delineate cell-type-specific responses and explore intercellular communication among different cell types in target vascular tissues. Comparative analysis of diseased versus control tissues helps elucidate the mechanisms underlying the onset and progression of vascular diseases. A major knowledge gap exists between iron dysregulation and PAD. While iron overload has been linked to oxidative stress and vascular dysfunction in atherosclerosis and aortic aneurysm, mechanistic studies in ischemic PAD skeletal muscle and microvasculature are limited. Future work is needed to define how iron handling and

**Table 3. Major Clinical Trials of Iron Supplementation in Heart Failure**

| Trial and year                                                    | Sample size | Design                                                           | Study drug                                                                                                                          | Inclusion criteria                                                                               | Primary end point                                                                                             | Outcome                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------|-------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FERRIC HF 2008 (NCT00125996) <sup>92</sup>                        | 35          | Randomized, controlled, observer-blinded trial                   | Iron sucrose (200 mg weekly until ferritin >500 ng/mL)                                                                              | CHF (NYHA II or III) with LVEF ≤45%                                                              | Change in peak VO <sub>2</sub> at 18 wk                                                                       | Improved exercise capacity                                                                                                                                                                                                                |
| FAIR-HF 2009 (NCT00520780) <sup>93</sup>                          | 459         | Randomized, double-blind, placebo-controlled                     | FCM 200 mg, IV, weekly up to iron repletion, then monthly until 24 wk                                                               | CHF (NYHA II–III), LVEF ≤40% (NYHA II) or ≤45% (NYHA III)                                        | Self-reported patient global assessment and NYHA functional class at week 24                                  | Improved symptoms and functional class                                                                                                                                                                                                    |
| CONFIRM HF 2014 (NCT01453608) <sup>94</sup>                       | 304         | Randomized, double-blind, placebo-controlled                     | FCM, IV, 200 mg, for 52 wk                                                                                                          | Stable ambulatory HFrEF, NYHA II or III, LVEF ≤45%                                               | Change in 6MWT distance at 24 wk                                                                              | Improvement in functional capacity over time                                                                                                                                                                                              |
| EFFECT-HF 2017 (NCT01394562) <sup>95</sup>                        | 172         | Randomized, open-label, endpoint-blinded                         | FCM, IV, 500 mg, 3 times                                                                                                            | HFrEF (LVEF ≤45%)                                                                                | Change in peak VO <sub>2</sub> at 24 wk                                                                       | Improved functional capacity                                                                                                                                                                                                              |
| AFFIRM-AHF 2020 (NCT02937454) <sup>96</sup>                       | 1132        | Randomized, double-blind, placebo-controlled                     | FCM, IV, up to 24 wk                                                                                                                | Patients admitted for acute HF with LVEF <50%                                                    | Recurrent HF hospitalizations and cardiovascular death up to 52 wk                                            | Reduced HF hospitalizations; no significant effect on cardiovascular death                                                                                                                                                                |
| IRONMAN 2022 (NCT02642562) <sup>97</sup>                          | 1137        | Randomized, open-label                                           | Ferric derisomaltose, 20 mg/kg IV, every 4 mo until trial completion                                                                | LVEF ≤45%, iron deficiency (serum ferritin <100 ng/mL or TSAT <20%)                              | HF hospitalization and cardiovascular outcomes over a median 2.7 y                                            | Fewer serious cardiac adverse events: trends toward reduced hospitalization. In the prespecified COVID-19 sensitivity analysis primary outcome, the RR was nominally statistically significant, which ultimately reduced the HF admission |
| PIVOTAL trial 2018 (EudraCT number, 2013-002267-25) <sup>98</sup> | 2141        | Randomized, multicenter, open-label, end point blinded           | High-dose iron sucrose, (400 mg IV monthly) or low-dose, (0–400 mg IV) monthly                                                      | Patients on maintenance hemodialysis receiving erythropoiesis-stimulating agents                 | Time to all-cause death or a composite of MI, stroke, and HF hospitalization                                  | High-dose intravenous iron reduced the risk of first and recurrent all-cause death, cardiovascular events, and HF events compared with low-dose iron                                                                                      |
| TACT 2013 (NCT00044213) <sup>99</sup>                             | 1708        | Double-blind, placebo-controlled, 2×2 factorial randomized trial | Disodium EDTA-based chelation therapy (40 intravenous infusions: weekly for 30 wk followed by 10 additional infusions 2–8 wk apart) | Adults aged ≥50 y with prior MI ≥6 wk before enrollment                                          | Composite of all-cause death, recurrent MI, stroke, coronary revascularization, or hospitalization for angina | In stable patients with a history of MI, EDTA chelation modestly reduced adverse cardiovascular events                                                                                                                                    |
| HEART-FID 2023 (NCT03037931) <sup>100</sup>                       | 3065        | Randomized, double-blind, placebo-controlled                     | FCM, 15 mg/kg IV for weighing <50 kg or 750 mg for weighing >50 kg, 2 injections (at 7-d intervals), repeated every 6 mo            | LVEF ≤40%, NYHA II–IV with iron deficiency (Ferritin <100 ng/mL or 100–300 ng/mL with TSAT <20%) | Death (12 mo), HF hospitalization (12 mo), change in 6MWD (6 mo)                                              | No apparent difference. Numerical improvements in death, HF hospitalization, and functional capacity                                                                                                                                      |

(Continued)

**Table 3. Continued**

| Trial and year                                    | Sample size | Design                                                    | Study drug                                                                                            | Inclusion criteria                                                                                                  | Primary end point                                                 | Outcome                                                                                      |
|---------------------------------------------------|-------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| FAIR-HF2<br>2025<br>(NCT03036462) <sup>101</sup>  | 1105        | Multicenter, randomized, double-blind, placebo-controlled | FCM, up to 2000 mg IV, followed by 500 mg every 4 mo                                                  | LVEF ≤45%, iron deficiency (serum ferritin level <100 ng/mL; or if TSAT <20%, a serum ferritin level 100–299 ng/mL) | Cardiovascular death and HF hospitalization (time to first event) | FCM was well tolerated but did not meet predefined statistical criteria for efficacy         |
| IRON-5 HF<br>2013<br>(NCT00386126) <sup>102</sup> | 23          | Randomized, double blind, placebo-controlled              | Ferrous sulfate (200 mg, PO 3 times daily, for 8 wk); Iron sucrose (200 mg, IV, once a week for 5 wk) | LVEF <40% NYHA II–IV                                                                                                | Variation of peak VO <sub>2</sub>                                 | Intravenous iron is likely superior in improving functional capacity compared with oral iron |
| IRONOUT<br>2016<br>(NCT02188784) <sup>103</sup>   | 225         | Randomized, double-blinded, placebo-controlled            | Polysaccharide iron complex, PO, 150 mg, twice daily, for 16 wk                                       | HFrEF, LVEF ≤40%, NYHA II–IV                                                                                        | Change in peak VO <sub>2</sub> at 16 wk                           | No significant improvement in exercise capacity                                              |

6MWT indicates 6-minute walking test; CHF, congestive heart failure; FCM, ferric carboxymaltose; HF, heart failure; HFrEF, heart failure with reduced ejection fraction; LVEF, left ventricular ejection fraction; MI, myocardial infarction; MWD, maximum walking distance; NYHA, New York Heart Association; and TSAT, transferrin saturation.

ferroptosis in ECs, VSMCs, and immune cells, and their intercellular interactions, contribute to disease severity and limb outcomes. Translational efforts can integrate iron-related molecular signatures with functional outcomes. Vascular organoids are powerful models for investigating vascular development, disease mechanisms, and regenerative medicine. They support the study of physiological and pathophysiological processes and provide a platform for exploring mechanisms and screening drugs. Moreover, clinical trials could assess whether correcting iron overload or deficiency improves vascular cell function and reduces recurrent cardiovascular events. Despite the encouraging results, additional studies and more preclinical research are needed to refine these strategies and reduce the risk of adverse effects associated with iron modulation. Integrating circulating biomarkers with imaging-based iron quantification may enhance trial design and enable more precise patient selection.

## CONCLUSIONS

Iron homeostasis is critical for maintaining cellular function and various physiological processes. Excessive or deficient iron is detrimental to living organisms. The pathogenic role of iron overload in vascular diseases has been widely recognized in recent years. Indeed, both iron overload and iron deficiency can impair vascular cell function, underscoring the indispensability of a balanced iron status for vascular homeostasis. Thus, the dual impacts of iron dysfunction and the underlying molecular mechanisms warrant further elucidation. Although animal models of iron overload or deficiency are used for in vivo studies,

how well current animal models can replicate human vascular pathology needs careful evaluation. Over the past decade, accumulating experimental data suggest that ferroptosis contributes to the pathophysiology of CVDs. However, the specific mechanisms underlying ferroptosis in various vascular diseases remain to be fully investigated. The significance of ferroptosis inhibition needs to be evaluated in clinical trials. Identification and validation of sensitive and specific biomarkers beyond conventional iron indices are needed for quantification of iron dysregulation, prognosis, and translational applications. Exploring the impact of iron dysregulation and underlying mechanisms will help identify potential therapeutic targets and develop novel approaches to treat CVD.

## ARTICLE INFORMATION

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None.



## REFERENCES

- Sawicki KT, De Jesus A, Ardehali H. Iron metabolism in cardiovascular disease: physiology, mechanisms, and therapeutic targets. *Circ Res*. 2023;132:379–396. doi: [10.1161/Circresaha.122.321667](https://doi.org/10.1161/Circresaha.122.321667)
- Wang H, Huang ZM, Du CY, Dong MQ, Fragasso G. Iron dysregulation in cardiovascular diseases. *Rev Cardiovasc Med*. 2024;25:16. doi: [10.31083/j.rcm2501016](https://doi.org/10.31083/j.rcm2501016)
- Rouault TA. The role of iron regulatory proteins in mammalian iron homeostasis and disease. *Nat Chem Biol*. 2006;2:406–414. doi: [10.1038/nchembio807](https://doi.org/10.1038/nchembio807)
- Leipuviene R, Theil EC. The family of iron responsive RNA structures regulated by changes in cellular iron and oxygen. *Cell Mol Life Sci*. 2007;64:2945–2955. doi: [10.1007/s00018-007-7198-4](https://doi.org/10.1007/s00018-007-7198-4)
- Brazzollotto X, Gaillard J, Pantopoulos K, Hentze MW, Moulis JM. Human cytoplasmic aconitase (iron regulatory protein 1) is converted into its [3Fe-4S] form by hydrogen peroxide in vitro but is not activated for iron-responsive element binding. *J Biol Chem*. 1999;274:21625–21630. doi: [10.1074/jbc.274.31.21625](https://doi.org/10.1074/jbc.274.31.21625)
- Wang H, Shi H, Rajan M, Canarie ER, Hong S, Simoneschi D, Pagano M, Bush MF, Stoll S, Leibold EA, et al. FBXL5 regulates IRP2 stability in iron homeostasis via an oxygen-responsive [2Fe2S] cluster. *Mol Cell*. 2020;78:e35. doi: [10.1016/j.molcel.2020.02.011](https://doi.org/10.1016/j.molcel.2020.02.011)
- Anderson SA, Nizzi CP, Chang YI, Deck KM, Schmidt PJ, Galy B, Damernsawad A, Broman AT, Kendziora C, Hentze MW, et al. The IRP1-HIF-2 $\alpha$  axis coordinates iron and oxygen sensing with erythropoiesis and iron absorption. *Cell Metab*. 2013;17:282–290. doi: [10.1016/j.cmet.2013.01.007](https://doi.org/10.1016/j.cmet.2013.01.007)
- Ghosh MC, Zhang DL, Jeong SY, Kovtunovych G, Ollivierre-Wilson H, Noguchi A, Tu T, Senecal T, Robinson G, Crooks DR, et al. Deletion of iron regulatory protein 1 causes polycythemia and pulmonary hypertension in mice through translational derepression of HIF2 $\alpha$ . *Cell Metab*. 2013;17:271–281. doi: [10.1016/j.cmet.2012.12.016](https://doi.org/10.1016/j.cmet.2012.12.016)
- Costain G, Ghosh MC, Maio N, Carnevale A, Si YC, Rouault TA, Yoon G. Absence of iron-responsive element-binding protein 2 causes a novel neurodegenerative syndrome. *Brain*. 2019;142:1195–1202. doi: [10.1093/brain/awz072](https://doi.org/10.1093/brain/awz072)
- Attieh ZK, Mukhopadhyay CK, Seshadri V, Tripoulas NA, Fox PL. Ceruloplasmin ferroxidase activity stimulates cellular iron uptake by a trivalent cation-specific transport mechanism. *J Biol Chem*. 1999;274:11116–11123. doi: [10.1074/jbc.274.2.1116](https://doi.org/10.1074/jbc.274.2.1116)
- Zeng W, Cai J, Zhang L, Peng Q. Iron deposition in Parkinson's disease: a mini-review. *Cell Mol Neurobiol*. 2024;44:26. doi: [10.1007/s10571-024-01459-4](https://doi.org/10.1007/s10571-024-01459-4)
- Niu L, Zhou Y, Lu L, Su A, Guo X. Ceruloplasmin deficiency impaired brain iron metabolism and behavior in mice. *Cell Biochem Biophys*. 2022;80:385–393. doi: [10.1007/s12013-022-01061-9](https://doi.org/10.1007/s12013-022-01061-9)
- Kremastinos DT, Farmakis D. Iron overload cardiomyopathy in clinical practice. *Circulation*. 2011;124:2253–2263. doi: [10.1161/CIRCULATIONAHA.111.050773](https://doi.org/10.1161/CIRCULATIONAHA.111.050773)
- Santos P, Krieger JE, Pereira AC. Molecular diagnostic and pathogenesis of hereditary hemochromatosis. *Int J Mol Sci*. 2012;13:1497–1511. doi: [10.3390/ijms13021497](https://doi.org/10.3390/ijms13021497)
- Kotze MJ, van Velden DP, Rensburg SJ, Erasmus R. Pathogenic mechanisms underlying iron deficiency and iron overload: new insights for clinical application. *EJIFCC*. 2009;20:108–123.
- Vinchi F, Porto G, Simmelbauer A, Altamura S, Passos ST, Garbowski M, Silva AMN, Spaich S, Seide SE, Spalla R, et al. Atherosclerosis is aggravated by iron overload and ameliorated by dietary and pharmacological iron restriction. *Eur Heart J*. 2020;41:2681–2695. doi: [10.1093/eurheartj/ehz112](https://doi.org/10.1093/eurheartj/ehz112)
- Ahmed S, Peterson SJ, Parikh MA, Frishman WH. Cardiovascular manifestations of hemochromatosis: a review of pathophysiology, mechanisms, and treatment options. *Cardiol Rev*. 2023;33:359–364. doi: [10.1097/CRD.0000000000000622](https://doi.org/10.1097/CRD.0000000000000622)
- Bacon BR, Adams PC, Kowdley KV, Powell LW, Tavill AS; American Association for the Study of Liver D. Diagnosis and management of hemochromatosis: 2011 practice guideline by the American Association for the Study of Liver Diseases. *Hepatology*. 2011;54:328–343. doi: [10.1002/hep.24330](https://doi.org/10.1002/hep.24330)
- Tavill AS; American Association for the Study of Liver D, American College of G, American Gastroenterological A. Diagnosis and management of hemochromatosis. *Hepatology*. 2001;33:1321–1328. doi: [10.1053/jhep.2001.24783](https://doi.org/10.1053/jhep.2001.24783)
- McLaren CE, Barton JC, Adams PC, Harris EL, Acton RT, Press N, Reboussin DM, McLaren GD, Sholinsky P, Walker AP, et al. Hemochromatosis and iron overload screening (HEIRS) study design for an evaluation of 100,000 primary care-based adults. *Am J Med Sci*. 2003;325:53–62. doi: [10.1097/00000441-200302000-00001](https://doi.org/10.1097/00000441-200302000-00001)
- Powell LW, George DK, McDonnell SM, Kowdley KV. Diagnosis of hemochromatosis. *Ann Intern Med*. 1998;129:925–931. doi: [10.7326/0003-4819-129-11\\_part\\_2-199812011-00002](https://doi.org/10.7326/0003-4819-129-11_part_2-199812011-00002)
- European Association for the Study of the L. EASL clinical practice guidelines on haemochromatosis. *J Hepatol*. 2022;77:479–502. doi: [10.1016/j.jhep.2022.03.033](https://doi.org/10.1016/j.jhep.2022.03.033)
- Esposito BP, Breuer W, Sirankapracha P, Pootrakul P, Herskho C, Cabantchik ZI. Labile plasma iron in iron overload: redox activity and susceptibility to chelation. *Blood*. 2003;102:2670–2677. doi: [10.1182/blood-2003-03-0807](https://doi.org/10.1182/blood-2003-03-0807)
- Aljwaid H, White DL, Collard KJ, Moody AJ, Pinkney JH. Non-transferrin-bound iron is associated with biomarkers of oxidative stress, inflammation and endothelial dysfunction in type 2 diabetes. *J Diabetes Complications*. 2015;29:943–949. doi: [10.1016/j.jdiacomp.2015.05.017](https://doi.org/10.1016/j.jdiacomp.2015.05.017)
- Orino K, Lehman L, Tsuji Y, Ayaki H, Torti SV, Torti FM. Ferritin and the response to oxidative stress. *Biochem J*. 2001;357:241–247. doi: [10.1042/0264-6021:3570241](https://doi.org/10.1042/0264-6021:3570241)
- Lakhal-Littleton S. Iron deficiency as a therapeutic target in cardiovascular disease. *Pharmaceuticals-Basel*. 2019;12:125. doi: [10.3390/ph12030125](https://doi.org/10.3390/ph12030125)
- Savarese G, von Haehling S, Butler J, Cleland JGF, Ponikowski P, Anker SD. Iron deficiency and cardiovascular disease. *Eur Heart J*. 2023;44:14–27. doi: [10.1093/eurheartj/ehac569](https://doi.org/10.1093/eurheartj/ehac569)
- Shanmugam NK, Chen K, Cherayil BJ. Commensal bacteria-induced interleukin 1 $\beta$  (IL-1 $\beta$ ) secreted by macrophages up-regulates hepcidin expression in hepatocytes by activating the bone morphogenetic protein signaling pathway. *J Biol Chem*. 2015;290:30637–30647. doi: [10.1074/jbc.M115.689190](https://doi.org/10.1074/jbc.M115.689190)
- Ross KL, Eisenstein RS. Iron deficiency decreases mitochondrial aconitase abundance and citrate concentration without affecting tricarboxylic acid cycle capacity in rat liver. *J Nutr*. 2002;132:643–651. doi: [10.1093/jn/132.4.643](https://doi.org/10.1093/jn/132.4.643)
- Institute of Medicine (US) Panel on Micronutrients. Chapter 9: iron. *Dietary reference intakes for vitamin A, vitamin K, arsenic, boron, chromium, copper, iodine, iron, manganese, molybdenum, nickel, silicon, vanadium, and zinc*. Washington, DC: National Academies Press; 2001:290–393. <https://www.ncbi.nlm.nih.gov/books/NBK222309/>
- Zimmermann MB, Hurrell RF. Nutritional iron deficiency. *Lancet*. 2007;370:511–520. doi: [10.1016/S0140-6736\(07\)61235-5](https://doi.org/10.1016/S0140-6736(07)61235-5)
- Ru Q, Li YS, Chen L, Wu YX, Min JX, Wang FD. Iron homeostasis and ferroptosis in human diseases: mechanisms and therapeutic prospects. *Signal Transduct Tar*. 2024;9:271. doi: [10.1038/s41392-024-01969-z](https://doi.org/10.1038/s41392-024-01969-z)
- Xu SW. Iron and atherosclerosis: the link revisited. *Trends Mol Med*. 2019;25:659–661. doi: [10.1016/j.molmed.2019.05.012](https://doi.org/10.1016/j.molmed.2019.05.012)
- Zhang WJ, Frei B. Intracellular metal ion chelators inhibit TNF $\alpha$ -induced SP-1 activation and adhesion molecule expression in human aortic endothelial cells. *Free Radic Biol Med*. 2003;34:674–682. doi: [10.1016/s0891-5849\(02\)01375-8](https://doi.org/10.1016/s0891-5849(02)01375-8)
- Duffy SJ, Biegelsen ES, Holbrook M, Russell JD, Gokce N, Keaney JF, Vita JA. Iron chelation improves endothelial function in patients with coronary artery disease. *Circulation*. 2001;103:2799–2804. doi: [10.1161/01.cir.103.23.2799](https://doi.org/10.1161/01.cir.103.23.2799)
- Guo Z, Ran Q, Roberts LJ, Zhou L, Richardson A, Sharan C, Wu D, Yang H. Suppression of atherogenesis by overexpression of glutathione peroxidase-4 in apolipoprotein E-deficient mice. *Free Radical Bio Med*. 2008;44:343–352. doi: [10.1016/j.freeradbiomed.2007.09.009](https://doi.org/10.1016/j.freeradbiomed.2007.09.009)
- Bai T, Li MX, Liu YF, Qiao ZT, Wang ZW. Inhibition of ferroptosis alleviates atherosclerosis through attenuating lipid peroxidation and endothelial dysfunction in mouse aortic endothelial cell. *Free Radical Bio Med*. 2020;160:92–102. doi: [10.1016/j.freeradbiomed.2020.07.026](https://doi.org/10.1016/j.freeradbiomed.2020.07.026)
- Gao F, Zhang B, Sun ZF, Gao Y, Liu CY, Dou XY, Tong HK, Wang R. Regulation of endothelial ferroptosis by SESN1 in atherosclerosis and its related mechanism. 2023;15:5052–5065. doi: [10.18632/aging.204777](https://doi.org/10.18632/aging.204777)
- Huang TC, Wang KJ, Li YW, Ye YC, Chen YX, Wang JF, Yao C. Construction of a novel ferroptosis-related gene signature of

- atherosclerosis. *Front Cell Dev Biol.* 2022;9:800833. doi: [10.3389/fcell.2021.800833](#)
40. Hennigar SR, Berryman CE, Harris MN, Karl JP, Lieberman HR, McClung JP, Rood JC, Pasiakos SM. Testosterone administration during energy deficit suppresses hepcidin and increases iron availability for erythropoiesis. *J Clin Endocrinol Metab.* 2020;105:dgz316. doi: [10.1210/clinem/dgz316](#)
  41. Xu H, Song Y, Xu J, Gu Y, Zhang Q, Liu L, Meng G, Wu H, Xia Y, Bao X, et al. Increased serum ferritin levels are independently associated with carotid atherosclerosis in women. *Br J Nutr.* 2017;117:1623–1630. doi: [10.1017/S0007114517001544](#)
  42. Fukami K, Yamagishi S, Iida S, Matsuoka H, Okuda S. Involvement of iron-evoked oxidative stress in smoking-related endothelial dysfunction in healthy young men. *PLoS One.* 2014;9:e89433. doi: [10.1371/journal.pone.0089433](#)
  43. Lu HC, Sun JJ, Hamblin MH, Chen YE, Fan YB. Transcription factor EB regulates cardiovascular homeostasis. *EBioMedicine.* 2021;63:103207. doi: [10.1016/j.ebiom.2020.103207](#)
  44. Li S, Zhang X. Iron in cardiovascular disease: challenges and potentials. *Front Cardiovasc Med.* 2021;8:707138. doi: [10.3389/fcvm.2021.707138](#)
  45. Wong CM, Preston IR, Hill NS, Suzuki YJ. Iron chelation inhibits the development of pulmonary vascular remodeling. *Free Radical Bio Med.* 2012;53:1738–1747. doi: [10.1016/j.freeradbiomed.2012.08.576](#)
  46. Fu S, Chen J, Liu B, Liang P, Zeng Y, Feng M, Xu Z, Zheng G, Yang S, Xu A, et al. Systemic inflammation modulates the ability of serum ferritin to predict all-cause and cardiovascular mortality in peritoneal dialysis patients. *BMC Nephrol.* 2020;21:237. doi: [10.1186/s12882-020-01892-9](#)
  47. Kawada S, Nagasawa Y, Kawabe M, Ohyama H, Kida A, Kato-Kogoe N, Nanami M, Hasuike Y, Kuragano T, Kishimoto H, et al. Iron-induced calcification in human aortic vascular smooth muscle cells through interleukin-24 (IL-24), with/without TNF- $\alpha$ . 2018;8:658. doi: [10.1038/s41598-017-19092-1](#)
  48. Song Y, Yang N, Si H, Liu T, Wang H, Geng H, Qin Q, Guo Z. Iron overload impairs renal function and is associated with vascular calcification in rat aorta. *Biomaterials.* 2022;35:1325–1339. doi: [10.1007/s10534-022-00449-7](#)
  49. Aierken Y, He H, Li R, Lin Z, Xu T, Zhang L, Wu Y, Liu Y. Inhibition of Slc39a14/Slc39a8 reduce vascular calcification via alleviating iron overload induced ferroptosis in vascular smooth muscle cells. *Cardiovasc Diabetol.* 2024;23:186. doi: [10.1186/s12933-024-02224-z](#)
  50. Sayadi A, Nguyen AT, Bard FA, Bard-Chapeau EA. Zip14 expression induced by lipopolysaccharides in macrophages attenuates inflammatory response. *Inflamm Res.* 2013;62:133–143. doi: [10.1007/s00011-012-0559-y](#)
  51. Sawada H, Hao H, Naito Y, Oboshi M, Hirotani S, Mitsuno M, Miyamoto Y, Hirota S, Masuyama T. Aortic iron overload with oxidative stress and inflammation in human and murine abdominal aortic aneurysm. *Arterioscler Thromb Vas.* 2015;35:1507–1514. doi: [10.1161/Atvbaha.115.305586](#)
  52. Da Ros F, Carnevale R, Cifelli G, Bizzotto D, Casaburo M, Perrotta M, Carnevale L, Vinciguerra I, Fardella S, Iacobucci R, et al. Targeting interleukin-1 $\beta$  protects from aortic aneurysms induced by disrupted transforming growth factor  $\beta$  signaling. *Immunity.* 2017;47:959. doi: [10.1016/j.immuni.2017.10.016](#)
  53. Ren JR, Lv YZ, Wu LL, Chen SL, Lei CX, Yang D, Li FD, Liu CZ, Zheng YH. Key ferroptosis-related genes in abdominal aortic aneurysm formation and rupture as determined by combining bioinformatics techniques. *Front Cardiovasc Med.* 2022;9:875434. doi: [10.3389/fcvm.2022.875434](#)
  54. He X, Xiong YC, Liu Y, Li YZ, Zhou HY, Wu KM. Ferrostatin-1 inhibits ferroptosis of vascular smooth muscle cells and alleviates abdominal aortic aneurysm formation through activating the SLC7A11/GPX4 axis. *FASEB J.* 2024;38:e23401. doi: [10.1096/fj.202300198RRR](#)
  55. Hu XR, Cai XY, Ma RS, Fu WW, Zhang CJ, Du XJ. Iron-load exacerbates the severity of atherosclerosis via inducing inflammation and enhancing the glycolysis in macrophages. *J Cell Physiol.* 2019;234:18792–18800. doi: [10.1002/jcp.28518](#)
  56. Zhang WJ, Wei H, Frei B. The iron chelator, desferrioxamine, reduces inflammation and atherosclerotic lesion development in experimental mice. *Exp Biol Med (Maywood).* 2010;235:633–641. doi: [10.1258/ebm.2009.009229](#)
  57. Malhotra R, Wunderer F, Barnes HJ, Bagchi A, Buswell MD, O'Rourke CD, Slocum CL, Ledsky CD, Peneyra KM, Sigurslid H, et al. Hepcidin deficiency protects against atherosclerosis. *Arterioscler Thromb Vas.* 2019;39:178–187. doi: [10.1161/Atvbaha.118.312215](#)
  58. Cai J, Zhang M, Liu Y, Li H, Shang L, Xu T, Chen Z, Wang F, Qiao T, Li K. Iron accumulation in macrophages promotes the formation of foam cells and development of atherosclerosis. *Cell Biosci.* 2020;10:137. doi: [10.1186/s13578-020-00500-5](#)
  59. Kautz L, Gabayan V, Wang X, Wu J, Onwuzurike J, Jung G, Qiao B, Lusis AJ, Ganz T, Nemeth E. Testing the iron hypothesis in a mouse model of atherosclerosis. *Cell Rep.* 2013;5:1436–1442. doi: [10.1016/j.celrep.2013.11.009](#)
  60. Demetz E, Tymoszyk P, Hilbe R, Volani C, Haschka D, Heim C, Auer K, Lener D, Zeiger LB, Pfeiffer-Obermair C, et al. The haemochromatosis gene Hfe and Kupffer cells control LDL cholesterol homeostasis and impact on atherosclerosis development. *Eur Heart J.* 2020;41:3949–3959. doi: [10.1093/eurheartj/ehaa140](#)
  61. Li YY, Zhou Q, Zhang K, Meng X. Iron overload and abdominal aortic aneurysm. *Rev Cardiovasc Med.* 2024;25:361. doi: [10.31083/j.rcm2510361](#)
  62. Recalcatti S, Locati M, Gammella E, Invernizzi P, Cairo G. Iron levels in polarized macrophages: regulation of immunity and autoimmunity. *Autoimmun Rev.* 2012;11:883–889. doi: [10.1016/j.autrev.2012.03.003](#)
  63. Kitagawa T, Kosuge H, Uchida M, Iida Y, Dalman RL, Douglas T, McConnell MV. RGD targeting of human ferritin iron oxide nanoparticles enhances in vivo MRI of vascular inflammation and angiogenesis in experimental carotid disease and abdominal aortic aneurysm. *J Magn Reson Imaging.* 2017;45:1144–1153. doi: [10.1002/jmri.25459](#)
  64. Krebs JR, Bellotti P, Ueland W, Valisno JAC, Joseph Manual Kollareth D, Sharma S, Su G, Hartman JB, Adithan A, Spinoso M, et al. Pharmacological inhibition of ferroptosis attenuates experimental abdominal aortic aneurysm formation. *Arterioscler Thromb Vasc Biol.* 2025;45:2053–2068. doi: [10.1161/ATVBAHA.125.323269](#)
  65. Kuzmicka W, Manda-Handzik A, Mroczek A, Cieloch A, Moskalik A, Demkow U, Wachowska M, Ciepiela O. Iron excess affects release of neutrophil extracellular traps and reactive oxygen species but does not influence other functions of neutrophils. *Immunol Cell Biol.* 2022;100:87–100. doi: [10.1111/imcb.12509](#)
  66. Warnatsch A, Ioannou M, Wang Q, Papayannopoulos V. Inflammation. Neutrophil extracellular traps license macrophages for cytokine production in atherosclerosis. *Science.* 2015;349:316–320. doi: [10.1126/science.aaa8064](#)
  67. Drechsler M, Megens RT, van Zandvoort M, Weber C, Soehnlein O. Hyperlipidemia-triggered neutrophilia promotes early atherosclerosis. *Circulation.* 2010;122:1837–1845. doi: [10.1161/CIRCULATIONAHA.110.961714](#)
  68. Yarosz EL, Ye C, Kumar A, Black C, Choi EK, Seo YA, Chang CH. Cutting edge: activation-induced iron flux controls CD4 T cell proliferation by promoting proper IL-2R signaling and mitochondrial function. *J Immunol.* 2020;204:1708–1713. doi: [10.4049/jimmunol.1901399](#)
  69. Xia X, Wu H, Chen Y, Peng H, Wang S. Ferroptosis of T cell in inflammation and tumour immunity. *Clin Transl Med.* 2025;15:e70253. doi: [10.1002/ctm2.70253](#)
  70. Schimmer S, Sridhar V, Satan Z, Grebe A, Saad M, Wagner B, Kahlert N, Werner T, Richter D, Dittmer U, et al. Iron improves the antiviral activity of NK cells. *Front Immunol.* 2024;15:1526197. doi: [10.3389/fimmu.2024.1526197](#)
  71. Bethel M, Annex BH. Peripheral arterial disease: a small and large vessel problem. *Am Heart J Plus.* 2023;28:100291. doi: [10.1016/j.ahjo.2023.100291](#)
  72. Vinchi F, Muckenthaler MU, Da Silva MC, Balla G, Balla J, Jeney V. Atherogenesis and iron: from epidemiology to cellular level. *Front Pharmacol.* 2014;5:94. doi: [10.3389/fphar.2014.00094](#)
  73. Zacharski LR, Chow BK, Howes PS, Shamayeva G, Baron JA, Dalman RL, Malenka DJ, Ozaki CK, Lavori PW. Reduction of iron stores and cardiovascular outcomes in patients with peripheral arterial disease - a randomized controlled trial. 2007;297:603–610. doi: [10.1001/jama.297.6.603](#)
  74. Schiano E, Cappello E, Cecere D, Pompeo F, Novellino E, Stornaiuolo M, Izzo M. Increased levels of circulating iron-albumin complexes in peripheral arterial disease patients. *Antioxidants (Basel).* 2023;12:12. doi: [10.3390/antiox12020503](#)
  75. Cartland SP, Stanley CP, Bursill C, Passam F, Figtree GA, Patel S, Loa J, Golledge J, Robinson DA, Aitken SJ, et al. Sex, endothelial cell functions, and peripheral artery disease. *Int J Mol Sci.* 2023;24:17439. doi: [10.3390/ijms242417439](#)

76. Agarvas AR, Kopf S, Lopes TJS, Atkins JL, Thalmann P, Fernández-Real JM, Nawroth P, Muckenthaler MU. Iron biomarkers predict peripheral artery disease in females. *Atherosclerosis*. 2025;402:119111. doi: [10.1016/j.atherosclerosis.2025](https://doi.org/10.1016/j.atherosclerosis.2025)
77. Batchelor EK, Kapitsinou P, Pergola PE, Kovesdy CP, Jalal DI. Iron deficiency in chronic kidney disease: updates on pathophysiology, diagnosis, and treatment. *J Am Soc Nephrol*. 2020;31:456–468. doi: [10.1681/Asn.2019020213](https://doi.org/10.1681/Asn.2019020213)
78. Paul BT, Manz DH, Torti FM, Torti SV. Mitochondria and iron: current questions. *Expert Rev Hematol*. 2017;10:65–79. doi: [10.1080/17474086.2016.1268047](https://doi.org/10.1080/17474086.2016.1268047)
79. Chennupati R, Solga I, Wischmann P, Dahlmann P, Celik FG, Pacht D, Sahin A, Yogathasan V, Hosen MR, Gerdes N, et al. Chronic anemia is associated with systemic endothelial dysfunction. *Front Cardiovasc Med*. 2023;10:1099069. doi: [10.3389/fcvm.2023](https://doi.org/10.3389/fcvm.2023)
80. Lee YN, Wang HH, Su CH, Lee HI, Chou YH, Hsieh CL, Liu WT, Shu KT, Chang KT, Yeh HI, et al. Deferoxamine accelerates endothelial progenitor cell senescence and compromises angiogenesis. *Aging-US*. 2021;13:21364–21384. doi: [10.18632/aging.203469](https://doi.org/10.18632/aging.203469)
81. Wilkinson T, Fletcher E, Ring A, Bradley C, Papoutsis E, Miserlis D, Smith RS, Bohannon WT, Pipinos II, Koutakis P. Deciphering the role of ferroptosis in the pathogenesis of peripheral artery disease myopathy. *Biol-Basel*. 2025;14:14. doi: [10.3390/biology14050537](https://doi.org/10.3390/biology14050537)
82. Desormais I, Aboyans V, Bura A, Constans J, Cambou JP, Messas E, Labrunie A, Lacroix P. Anemia, an independent predictive factor for amputation and mortality in patients hospitalized for peripheral artery disease. *Eur J Vasc Endovasc*. 2014;48:202–207. doi: [10.1016/j.ejvs.2014.04.005](https://doi.org/10.1016/j.ejvs.2014.04.005)
83. Otaki Y, Watanabe T, Takahashi H, Sugai T, Yokoyama M, Tamura H, Kato S, Nishiyama S, Arimoto T, Shishido T, et al. Impact of iron deficiency on peripheral artery disease after endovascular therapy. *Circ Rep*. 2019;1:187–195. doi: [10.1253/circrep.CR-18-0029](https://doi.org/10.1253/circrep.CR-18-0029)
84. Esteban C, Hernández-Rodríguez I. Peripheral arterial disease and anaemia. *Med Clin-Barcelona*. 2022;158:221–228. doi: [10.1016/j.medcli.2021.07.010](https://doi.org/10.1016/j.medcli.2021.07.010)
85. Birmipili P, Cromwell LA, Li QJ, Johal AS, Atkins E, Waton S, Pherwani AD, Williams R, Richards T, Nandhra S. The impact of pre-operative anaemia on one year amputation free survival and re-admissions in patients undergoing vascular surgery for peripheral arterial disease: a National Vascular Registry Study. *Eur J Vasc Endovasc*. 2023;66:204–212. doi: [10.1016/j.ejvs.2023.05.003](https://doi.org/10.1016/j.ejvs.2023.05.003)
86. Gill D, Monori G, Tzoulaki I, Dehghan A. Iron status and risk of stroke. *Stroke*. 2018;49:2815–2821. doi: [10.1161/STROKEAHA.118.022701](https://doi.org/10.1161/STROKEAHA.118.022701)
87. Reinhold J, Papadopoulos C, Baral R, Vassiliou VS. Iron deficiency for prognosis in acute coronary syndrome - a systematic review and meta-analysis. *Int J Cardiol*. 2021;328:46–54. doi: [10.1016/j.ijcard.2020.12.021](https://doi.org/10.1016/j.ijcard.2020.12.021)
88. Ciacciarelli A, Falcou A, Nicolini E, Broccolini A, Frisullo G, Abruzzese S, Scala I, Anticoli S, Testani E, Montinaro E, et al. The prognostic role of iron deficiency in acute ischemic stroke patients: a prospective multicentric cohort study. *J Neurol Sci*. 2025;469:123371. doi: [10.1016/j.jns.2024.123371](https://doi.org/10.1016/j.jns.2024.123371)
89. Premawardhana A, Perera C, Wijethilaka MN, Wanasinghe SK, Rajakaruna R, Samarasinghe R, Williams S, Mettananda S. Efficacy and safety of deferoxamine, deferriox and deferiprone triple iron chelator combination therapy for transfusion-dependent beta-thalassaemia with very high iron overload: a protocol for randomised controlled clinical trial. *BMJ Open*. 2024;14:e077342. doi: [10.1136/bmjopen-2023-077342](https://doi.org/10.1136/bmjopen-2023-077342)
90. Li J, Xu L, Zuo YX, Chang XQ, Chi HT. Potential intervention target of atherosclerosis: ferroptosis (review). *Mol Med Rep*. 2022;26:343. doi: [10.3892/mmr.2022.12859](https://doi.org/10.3892/mmr.2022.12859)
91. Feng Y, Madungwe NB, Imam Aliagan AD, Tombo N, Bopassa JC. Liproxstatin-1 protects the mouse myocardium against ischemia/reperfusion injury by decreasing VDAC1 levels and restoring GPX4 levels. *Biochem Biophys Res Commun*. 2019;520:606–611. doi: [10.1016/j.bbrc.2019.10.006](https://doi.org/10.1016/j.bbrc.2019.10.006)
92. Okonko DO, Grzeslo A, Witkowski T, Mandal AK, Slater RM, Roughton M, Foldes G, Thum T, Majda J, Banasiak W, et al. Effect of intravenous iron sucrose on exercise tolerance in anemic and nonanemic patients with symptomatic chronic heart failure and iron deficiency FERRIC-HF: a randomized, controlled, observer-blinded trial. *J Am Coll Cardiol*. 2008;51:103–112. doi: [10.1016/j.jacc.2007.09.036](https://doi.org/10.1016/j.jacc.2007.09.036)
93. Anker SD, Comin Colet J, Filippatos G, Willenheimer R, Dickstein K, Drexler H, Luscher TF, Bart B, Banasiak W, Niegowska J, et al. Ferric carboxymaltose in patients with heart failure and iron deficiency. *N Engl J Med*. 2009;361:2436–2448. doi: [10.1056/NEJMoa0908355](https://doi.org/10.1056/NEJMoa0908355)
94. Ponikowski P, van Veldhuisen DJ, Comin-Colet J, Ertl G, Komajda M, Mareev V, McDonagh T, Parkhomenko A, Tavazzi L, Levesque V, et al. Beneficial effects of long-term intravenous iron therapy with ferric carboxymaltose in patients with symptomatic heart failure and iron deficiency. *Eur Heart J*. 2015;36:657–668. doi: [10.1093/eurheartj/ehu385](https://doi.org/10.1093/eurheartj/ehu385)
95. van Veldhuisen DJ, Ponikowski P, van der Meer P, Metra M, Bohm M, Doletsky A, Voors AA, Macdougall IC, Anker SD, Roubert B, et al. Effect of ferric carboxymaltose on exercise capacity in patients with chronic heart failure and iron deficiency. *Circulation*. 2017;136:1374–1383. doi: [10.1161/CIRCULATIONAHA.117.027497](https://doi.org/10.1161/CIRCULATIONAHA.117.027497)
96. Ponikowski P, Kirwan BA, Anker SD, McDonagh T, Dorobantu M, Drozd J, Fabien V, Filippatos G, Gohring UM, Keren A, et al. Ferric carboxymaltose for iron deficiency at discharge after acute heart failure: a multicentre, double-blind, randomised, controlled trial. *Lancet*. 2020;396:1895–1904. doi: [10.1016/S0140-6736\(20\)32339-4](https://doi.org/10.1016/S0140-6736(20)32339-4)
97. Kalra PR, Cleland JGF, Petrie MC, Thomson EA, Kalra PA, Squire IB, Ahmed FZ, Al-Mohammad A, Cowburn PJ, Foley PWX, et al. Intravenous ferric derisomaltose in patients with heart failure and iron deficiency in the UK (IRONMAN): an investigator-initiated, prospective, randomised, open-label, blinded-endpoint trial. *Lancet*. 2022;400:2199–2209. doi: [10.1016/S0140-6736\(22\)02083-9](https://doi.org/10.1016/S0140-6736(22)02083-9)
98. Macdougall IC, White C, Anker SD, Bhandari S, Farrington K, Kalra PA, McMurray JJV, Murray H, Tomson CRV, Wheeler DC, et al. Intravenous iron in patients undergoing maintenance hemodialysis. *N Engl J Med*. 2019;380:447–458. doi: [10.1056/NEJMoa1810742](https://doi.org/10.1056/NEJMoa1810742)
99. Lamas GA, Goertz C, Boineau R, Mark DB, Rozema T, Nahin RL, Lindblad L, Lewis EF, Drisko J, Lee KL. Effect of disodium EDTA chelation regimen on cardiovascular events in patients with previous myocardial infarction: the TACT randomized trial. *JAMA*. 2013;309:1241–1250. doi: [10.1001/jama.2013.2107](https://doi.org/10.1001/jama.2013.2107)
100. Mentz RJ, Garg J, Rockhold FW, Butler J, De Pasquale CG, Ezekowitz JA, Lewis GD, O'Meara E, Ponikowski P, Troughton RW, et al. Ferric carboxymaltose in heart failure with iron deficiency. *N Engl J Med*. 2023;389:975–986. doi: [10.1056/NEJMoa2304968](https://doi.org/10.1056/NEJMoa2304968)
101. Anker SD, Friede T, Butler J, Talha KM, Placzek M, Diek M, Nosko A, Stas A, Kluge S, Jarczok D, et al. Intravenous ferric carboxymaltose in heart failure with iron deficiency: the FAIR-HF2 DZHK05 randomized clinical trial. *JAMA*. 2025;333:1965–1976. doi: [10.1001/jama.2025.3833](https://doi.org/10.1001/jama.2025.3833)
102. Beck-da-Silva L, Piardi D, Soder S, Rohde LE, Pereira-Barretto AC, de Albuquerque D, Bocchi E, Vilas-Boas F, Moura LZ, Montera MW, et al. IRON-HF study: a randomized trial to assess the effects of iron in heart failure patients with anemia. *Int J Cardiol*. 2013;168:3439–3442. doi: [10.1016/j.ijcard.2013.04.181](https://doi.org/10.1016/j.ijcard.2013.04.181)
103. Lewis GD, Malhotra R, Hernandez AF, McNulty SE, Smith A, Felker GM, Tang WHW, LaRue SJ, Redfield MM, Semigran MJ, et al. Effect of oral iron repletion on exercise capacity in patients with heart failure with reduced ejection fraction and iron deficiency: the IRONOUT HF randomized clinical trial. *JAMA*. 2017;317:1958–1966. doi: [10.1001/jama.2017.5427](https://doi.org/10.1001/jama.2017.5427)
104. Zou C, Liu X, Xie R, Bao Y, Jin Q, Jia X, Li L, Liu R. Deferiprone attenuates inflammation and myocardial fibrosis in diabetic cardiomyopathy rats. *Biochem Biophys Res Commun*. 2017;486:930–936. doi: [10.1016/j.bbrc.2017.03.127](https://doi.org/10.1016/j.bbrc.2017.03.127)